Statistics Lecture 6 (Practical Session)

College of Computer Science and Information Technology

Software Department

The second stage

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correlation coefficient

The **correlation coefficient** is a statistical measure that describes the strength and direction of a relationship between two variables. It quantifies how closely the two variables move together.

The correlation coefficient denoted as r, and it ranges from -1 to 1:

- **r = 1**: A perfect positive correlation (as one variable increases, the other increases proportionally).
- **r = -1**: A perfect negative correlation (as one variable increases, the other decreases proportionally).
- **r = 0**: No correlation (the variables do not show any linear relationship).

Values between -1 and 1 indicate varying degrees of positive or negative correlations:

- **0 < r < 1**: Positive correlation (as one variable increases, the other tends to increase).
- **-1 < r < 0**: Negative correlation (as one variable increases, the other tends to decrease).

A higher absolute value of **r** indicates a stronger relationship. For example, **r = 0.9** suggests a strong positive relationship, while **r = -0.3** indicates a weak negative relationship.

Types of correlation

1-Bivariate Linear Correlation

Bivariate Linear Correlation measures the strength and direction of the linear relationship between two variables. In SPSS, you can compute this using **Pearson's correlation**, **Spearman's correlation**, or **Kendall's tau**.

Pearson Correlation (r)

- **Measures**: Linear relationship between two continuous (Quantitative) variables.
- **Range**: -1 to 1
- **When to Use**: When both variables are **normally distributed** and have a **linear relationship**.
- **Example**: The relationship between **height** and **weight**.

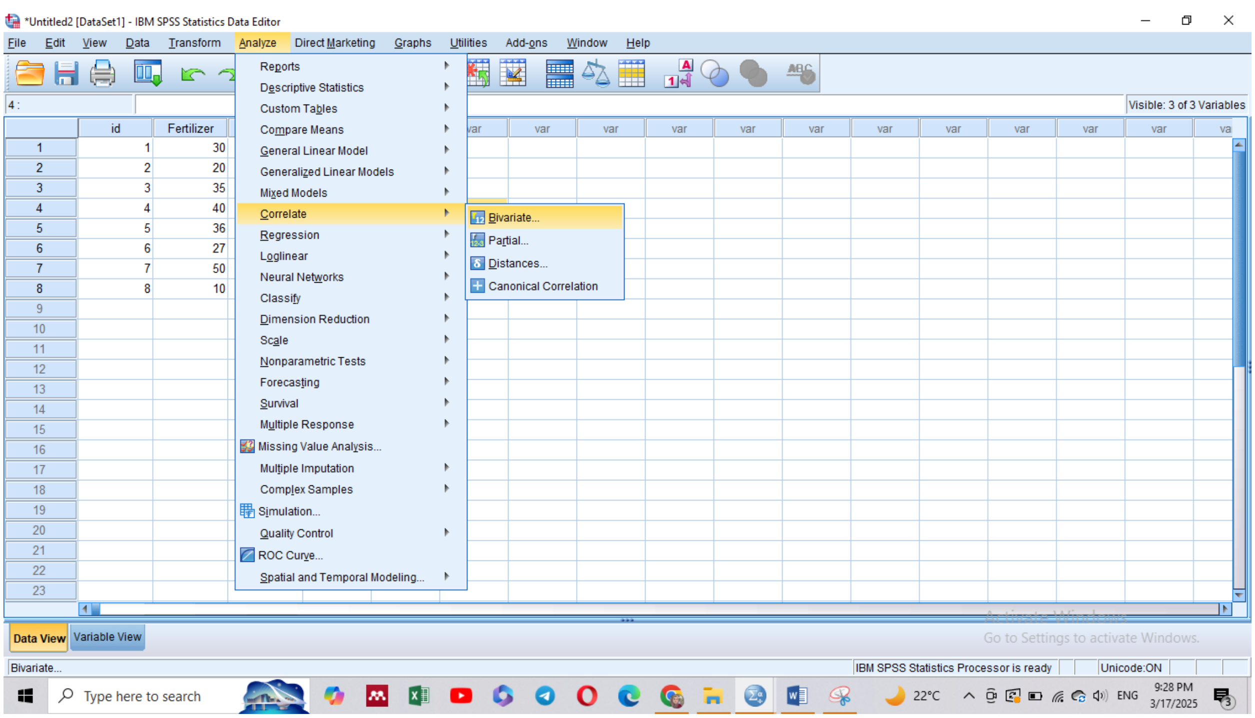
- **SPSS Option**: **Analyze → Correlate → Bivariate → Pearson**

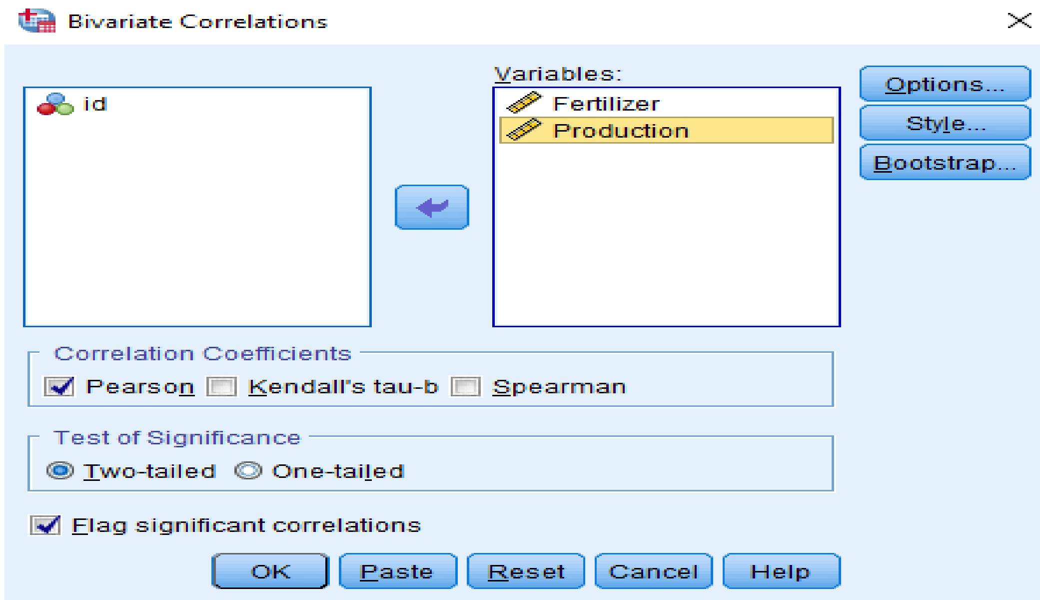
Example: Calculating Pearson correlation in spss

id	Fertilizer	Production
1	30	500
2	20	450
3	35	700
4	40	650
5	36	540
6	27	440
7	50	670
8	10	290

Step 1: We define the variables in the Variable Viewer interface and then enter the data in the Data Viewer interface

Step 2: Calculate Pearson correlation





Correlations

[DataSet1]

Correlations

		Fertilizer	Production
Fertilizer	Pearson Correlation	1	.897**
	Sig. (2-tailed)		.003
	N	8	8
Production	Pearson Correlation	.897**	1
	Sig. (2-tailed)	.003	
	N	8	8

**. Correlation is significant at the 0.01 level (2-tailed).

تدل قيمة معامل الارتباط بيرسون
0.897 على أن هنالك علاقة ارتباط
قوية بين كمية الإنتاج وكمية السماد

Spearman's Correlation (ρ)

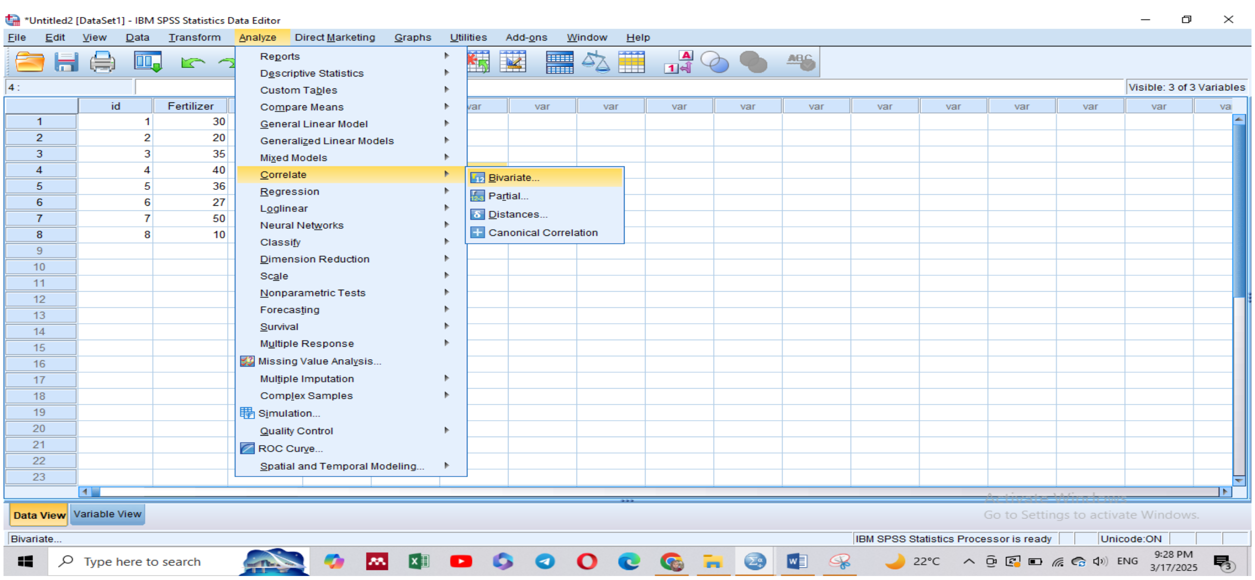
- **Measures:** The monotonic relationship (whether an increase in one variable generally leads to an increase/decrease in another).
- **Range:** -1 to 1
- **When to Use:** When data is ordinal (ranked) or not normally distributed.
- **Example:** The relationship between customer satisfaction ranking and product sales.
- **SPSS Option: Analyze → Correlate → Bivariate → Spearman**

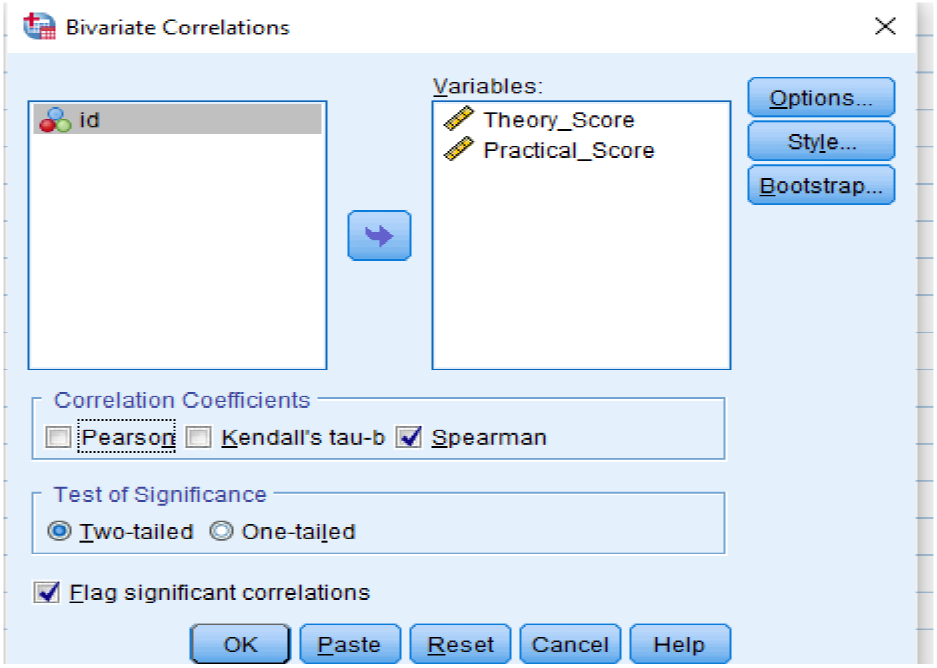
Example: Calculating Spearman's Correlation in spss

id	Theory_Sc...	Practical_Score
1	10	9
2	10	10
3	15	7
4	12	5
5	9	6
6	5	10
7	8	9
8	11	8
9	4	3
10	7	4

Step 1: We define the variables in the Variable Viewer interface and then enter the data in the Data Viewer interface

Step 2: Calculate **Spearman's Correlation**





Nonparametric Correlations

Correlations

			Theory_Score	Practical_Score
Spearman's rho	Theory_Score	Correlation Coefficient	1.000	.738*
		Sig. (2-tailed)	.	.015
		N	10	10
	Practical_Score	Correlation Coefficient	.738*	1.000
		Sig. (2-tailed)	.015	.
		N	10	10

*. Correlation is significant at the 0.05 level (2-tailed).

Kendall's Tau (τ)

- **Measures:** Similar to Spearman's but better for small sample sizes or when there are many tied ranks.
- **Range:** -1 to 1
- **When to Use:** When ranking is important, especially for small datasets.
- **Example:** The relationship between ranking in a race and finishing time.

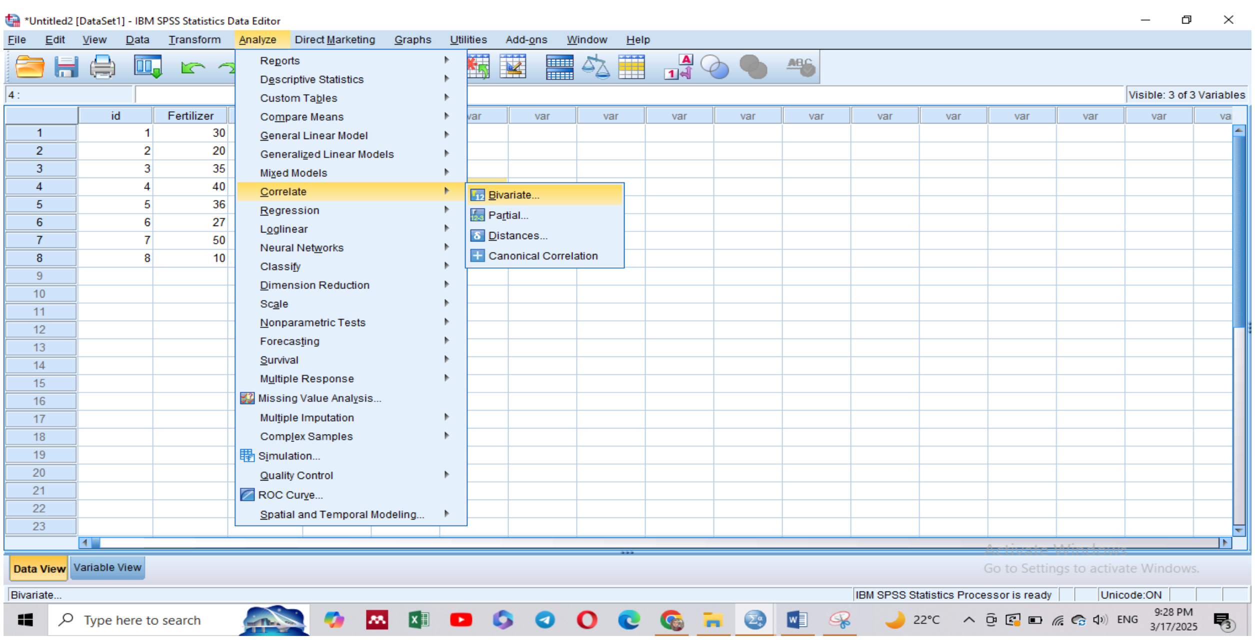
- **SPSS Option: Analyze → Correlate → Bivariate → Kendall’s Tau**
-

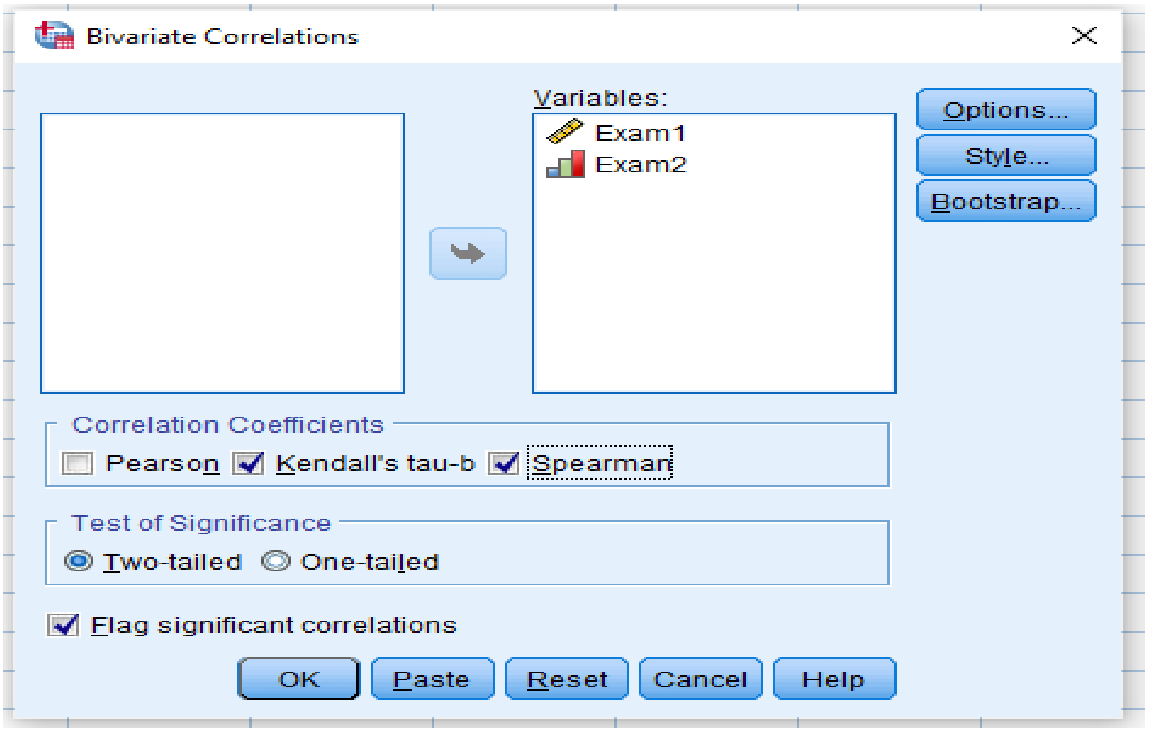
Example: Calculating Kendall’s Tau Correlation in spss

Step 1: We define the variables in the Variable Viewer interface and then enter the data in the Data Viewer interface

Step 2: Calculate **Kendall’s Tau Correlation**

Exam1	Exam2
85	85
98	95
90	80
83	75
57	70
63	65
77	73
99	93
80	79
96	88
69	74





Nonparametric Correlations

Correlations

			Exam1	Exam2
Kendall's tau_b	Exam1	Correlation Coefficient	1.000	.818**
		Sig. (2-tailed)	.	.000
		N	11	11
	Exam2	Correlation Coefficient	.818**	1.000
		Sig. (2-tailed)	.000	.
		N	11	11
Spearman's rho	Exam1	Correlation Coefficient	1.000	.955**
		Sig. (2-tailed)	.	.000
		N	11	11
	Exam2	Correlation Coefficient	.955**	1.000
		Sig. (2-tailed)	.000	.
		N	11	11

**. Correlation is significant at the 0.01 level (2-tailed).